Who governs the robots? A short brief (v5)

A short brief (v4, after a second collection round and a second adversarial review)

Centre for Enterprise Innovation (CEI) AI Robotics project, with Claude Code

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Version 5 note (collection round 3, 2 November 2026). The tables and figures in this package were regenerated after round 3, which recovered the text of round-2 instruments that could not be read before, harvested new service-robot, care, exoskeleton and humanoid instruments, re-read the truncated statutes in full and added supplementary fields (whom a rule binds, dates, legal effect of standards, robot-provision year). The narrative below still states the v4 counts of 31 October 2026 and has not been re-audited sentence by sentence against the v5 tables; where a number in the text and a number in a table differ, the table is current. The round-3 companion report (collection_round3) gives the v5 headline counts and what changed, and HUMAN_ASSISTANCE.md lists what still needs a person.

Counted as whole laws and regulations rather than as every legal text, 195 rules with legal force reach robots in this collection, and only 15 of them name AI. An *instrument* is any official text meant to steer conduct, from a statute to a voluntary guideline. A *binding* instrument has legal force; standards and guidelines do not. A *principal* instrument is a law or regulation counted once with its amendments, without its paperwork (commencement orders, fee rules, exemptions, corrigenda, EEA incorporation decisions, tables of contents, duplicates and stubs under 150 words). The *embodied set* is the documents the model rated as about AI in a body (robots, drones, vehicles) or tagged with a robot type. It holds 505 principal instruments, 195 of them binding. Of those, 168 are rated as genuinely about robots, 140 contain any provision on autonomy or AI, and 15 name AI. Drone rules (107) and automated-vehicle rules (67) make up most of them. Version 2 of this study reported five binding rules, version 3 reported 299, and neither jump happened in the world. Both reflect where the teams searched: version 2 looked on the web for texts about robots and missed the aviation, vehicle and machinery registers where binding law lives; version 3 counted every text harvested there, fee rules included. The first number was wrong, the second inflated.

The second collection round changed conclusions, not just counts. After eight regional teams' web searches, a second round sent AI search agents and database scripts into legal registers, standards catalogues and regulator sites and supplied 190 of the 195 binding instruments. The collectors, like the reviewers of every version, were AI agents (Claude) in one Claude Code session; every label came from an AI model (Claude Sonnet 5.5), and no person read or labelled any document. Version 3 said robot rules discuss fairness, transparency and privacy less than general-AI rules. Within the first round alone, none of the 14 core values differs between robot and general-AI instruments; within the second, 10 of 14 are lower, so the deficit belongs to the second round's short legal texts and is withdrawn. What holds in both rounds is the body: physical safety (57% of robot against 7% of general-AI first-round instruments), liability for damage, closeness to people and how much autonomy a machine may keep.

Scientists study companion robots; lawmakers regulate things that move. Companion and social robots are the subject of 43% of the 509 research papers but of 15 of the 505 principal instruments, none binding. Care robots and humanoids come next in research; humanoids have no binding instrument, care robots four. Drones are the mirror image: 6% of the

research, 42% of the instruments. Factory arms and collaborative robots roughly match; every gap named above is wider than chance would produce.

Everyone raises the control question; one binding rule in four answers it. "How much may the machine decide, and when must a person step in?" is raised by 61% of the 195 binding robot rules and set out in detail by 25%. Physical safety is what binding rules do detail (62%). Risk tiers and pre-market conformity checks grow more common each year, within the first round alone and across both.

What it means for you. A factory worker beside a collaborative robot is covered by ISO 10218 (safety requirements for robots working beside people) and in Europe by the Machinery Regulation 2023/1230 (conformity assessment and CE marking before sale); the 13 binding instruments reaching factory arms are machinery and workplace law, none written for robots. Someone buying a companion robot for a parent finds no binding rule written for it; what applies is the EU AI Act (bans on manipulative systems; a duty to tell people they face a machine), product-safety and data-protection law and the voluntary care-robot standard ISO 13482. A city buying delivery robots meets US state "personal delivery device" statutes, which admit them to sidewalks under weight, speed and insurance conditions, and Japan's Road Traffic Act, which treats small remote-operated vehicles as pedestrians after notice to the police.

Read with care. All counts are within this collection: 3,165 documents had enough text to label, 42% are European, 75% English, and 2026 is partial. On 160 second-round documents a second model from the same company (Claude Opus 5.5) agreed with the labels at a median kappa of 0.86, as in the first round; page quality is the weakest field. An expert audit of 159 risk-stratified version-3 rows found errors in 68% (62% material), now corrected; a cleaning fault that had deleted most of the text of Chinese, Japanese and Korean documents was fixed and those documents re-labelled. Even 195 is a ceiling: of 100 version-3 texts re-read independently, 54% were distinct, binding and substantively about robots. Most technical standards were read from scope pages only. Of 39 well-known instruments, 22 are captured with text; Germany's Autonomous Driving Act, Korea's Intelligent Robots Act, the EU Product Liability Directive and ISO/TS 15066 are missing, and Hangzhou's robot statute and Hong Kong's automated-vehicle regulation sit on hosts that blocked the collectors and head a third round. Also withdrawn from version 3: declines in attention to closeness to people and liability, a regional divergence on values (it appears among instruments, where it tracks the kinds of instrument each region's databases supply, and not among research papers), the El Salvador "only robot-titled statute" line and a rising binding share and jurisdiction count, which follow from the search design. The labels remain unvalidated by people.